R.K.D.F. UNIVERSITY, BHOPAL

M.TECH **(Construction Technology and Management)**

Second Year **(Semester – 3)**

Course Code:

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Construction Technology and Managment)	**Advanced Highway Construction**	**MTCTM-3001(A)**	**3L-1T-0P**	**4**

Courses Objectives

CO1:- Student will be learning technicality of selection Soil and its requirement in highway construction.

CO2:-Student will be learning various technical aspects of Bituminous construction in highway.

CO3:- Student will be learning various technical aspects of Cement Concrete Road construction in highway.

CO4:- Student will be learning various technical aspects of Reinforced Cement Concrete Road construction in highway.

CO5:- Student will be learning various tool and technic of project planning and management in highway construction

UNIT-1 Earthwork and Soling

Classification of types of highway construction, Suitability of each type under Indian conditions. Selection of base course and surface course. Selection of soils, construction of embankments, excavation and compaction equipments. Field and laboratory tests for quality control. Stone soling, brick soling, current practices. Construction of earth roads, gravel roads, soil stabilized roads, water bound macadam. Paved roads (i)bricks (ii)stones.

UNIT-2 Bituminous Construction

Properties, requirements and specifications of materials, equipments and plants. Detailed construction procedure of each type. Field and laboratory tests for quality control. Choice of binders under different conditions. IRC, British, and MOST Specifications. Bituminous surface treatments, interface treatments, prime coat, and tack coat, surface dressing and seal coat, grouted or penetration macadam, bituminous bound macadam, Sheet asphalt, bituminous concrete, mastic asphalt, dense tar surfacing.

UNIT-3 Cement Concrete Road Construction

Necessity of providing a base course under cement concrete road construction. Selection of materials, constructions methods, detailed construction procedure, Quality control tests(Lab. and Field).

Construction equipments.Classificationofvarioustypesofjoints,necessityofprovidingeachtype, method of construction of joints, load transfer devices, dowel bars, tie bars. joints filler and sealer materials, IRC Specifications.

UNIT-4 Reinforced Cement Concrete Road Construction

Necessity of providing reinforcement in cement concrete pavements, continuously reinforced concrete pavements, prestressed concrete pavements and fibre reinforced concrete pavements. Selection of themix, compaction method and construction procedure for each type. Recommendations under Indian conditions.

UNIT-5

Construction Planning and Management CPM/PERT in Highway Construction.

Reference

1. Highway Engineering: Planning Design, and Operation by Bastian J.Schroeder, Christopher Cunningham, Daniel J Findley and Tom Brown

2. Highway Engineering; Pavements, Materials and ControlofQuality by Athanassios Nikolaides

3. The Handbook Of Highway Engineering byT.F.Fwa

4. Traffic and Highway Engineering byLesterA.Hoel

5. Principle of Highway Engineering and TrafficEngineering byWilley

R.K.D.F. UNIVERSITY, BHOPAL

M.TECH **(Construction Technology and Management)**

Second Year (**Semester – 3)**

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Construction Technology and Managment)	**Multi Storied Buildings**	**MTCTM-3001(B)**	**3L-1T-0P**	**4**

Courses Objectives

CO1:- Student will be learning systematic planning of multi storied building

CO2:-Student will be learning various technical Multistoried Buildings, Preliminery design

CO3:- Student will be learning various technical aspects Analysis of Shear Walled Buildings

CO4:- Student will be learning various technical aspects of Yield line Analysis of reinforced concrete slabs

CO5:- Student will be learning technical aspects on Foundation-Superstructure interaction, Earthquake effects and design for ductility.

UNIT-1

Structural systems and their suitability, structural design criteria in planning.

UNIT-2Multistoried Buildings, Preliminery design, Analysis of building frames for vertical and lateral loads by approximate method, Matrix methods for the analysis of building frames & computer programming for the same.

UNIT-3 Analysis of Shear Walled Buildings Design of sections in reinforced concrete by working stress and limit state methods, Detailing of joints.

UNIT-4

Yield line Analysis of reinforced concrete slabs, concept to moment distribution.

UNIT-5

Foundation-Superstructure interaction, Earthquake effects and design for ductility.

Reference

Structural Design of MultiStoreyed Buildings byU.H.Varyani

Illustrative Design of ReinforcedBuilding byDr.S.R.Karve&Dr.V.L.Shah

Practical Handbook on BuildingConstruction byEr.M.K.Gupta

Civil Engineering Drawing&House Planning byB.P.Verma

Structral Analysis Of Regular MultiStorey Buildings byKarolyA.Zalka

R.K.D.F. UNIVERSITY, BHOPAL

M.TECH (Construction Technology and Management)

Second Year (Semester – 3)

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Construction Technology and Managment)	**Advanced Dam Design and Construction**	**MTCTM-3001(C)**	**3L-1T-0P**	**4**

Course Outcomes

After studying this subject, the students will be able to:

CO1: Understand the design and construction of gravity dams

CO2: Understand the design and construction of spillways

CO3: Understand the Elementary Design of Arch Dams

CO4: Understand the design and construction of earth dam.

CO5: Understand the Application of Photo elasticity to the Design of Dams

UNIT-1 Gravity Dams

River valley projects and their purpose, preliminary investigations and surveys, Selection of site for a reservoir; Types of Dams and their choice. Stability factors; Stresses, Elementary profile, low and high Dams, Forces acting on a Dam. Evolution of the profile of a Dam by Method of Zones, Practical profiles. Design of openings in Gravity Dams, contraction joints. Foundation treatment by Grouting.

UNIT-2 Spillways

Design of ogee spillway section, Bucket and Energy Dissipation arrangements: Design and Details of siphon, Shaft, side channel, and chute spillways, Miscellaneous types of spillways. Design of spillway crest gates and sluice gates, hoisting Machines.

UNIT-3 Elementary Design of Arch Dams

Definition of an Arch Dam, classification of Arch Dams. Principles of Elastic Theory and applied Trial Load Analysis, Inclined arches, Dome-Dams, Details and Methods of analysis.

UNIT-4 Earth Dams

Introduction, Design criteria, against over topping, Control of seepage, Theory of flow nets for homogeneous and Zoned embankments. Pore pressure, Stability of slopes, Methods of Analysis, slip circle Method, Protection of slopes, Protection against free passage of water, Rockfill dams.

UNIT-5Application of Photo elasticity to the Design of Dams.

Use of the Electrical Analogy Method in the Design of Dams, stress computations with embedded Electrical Instruments. River Diversion for construction of Dams, Constructional aspects in the Execution of River Valley project.

Reference

1. Advance Dam Engineering by Robert B.Jansen
2. Demand Appurtenant Hydraulic Structure by Ljubomir Tanchev
3. Hydraulic of Demand Reservoir by FuatSenturk
4. Hydraulic Structure by P.Novak, A.Moffat, CNalluri, R.Narayanan
5. Demand Public Safety by Robert B Jansen

R.K.D.F. UNIVERSITY, BHOPAL

M.TECH **(Construction Technology and Management)**

Second Year (**Semester – 3)**

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Construction Technology and Managment)	**Advanced Foundation Engineering**	**MTCTM-3001(D)**	**3L-1T-0P**	**4**

UNIT-1 Shallow Foundations
BearingCapacity,Terzaghisanalysis,Computationsofbearingcapacityfactors.Skempton's analysis. Meyerhof's analysis. Balla's theory. Hansen's theory.Design of Shallow Foundations.

UNIT-2 PileFoundation
Use of piles, Types of piles, Design of Piles, Group action in cohesive and cohesion less soils. Negativeskin fricton. Laterally loaded piles. Piles under inclined loads, pile load test, Hrennikoff Method.

UNIT-3 Engineering with Geosythetics
Introduction Basic Mechanism of reinforced earth strength characteristics of reinforced soil.

UNIT-4 Bridge Substructures
Introduction, elements of bridge substructure, stability analysis of well foundation, design of pie & abutments, sinking of wells.

UNIT-5 MarineSubstructures
Introduction, Types of Marine structures elements, design criteria, design of gravity wall, piled wharf structure breakwaters.

Reference

1. Advance Foundation Engineering by V.N.S.Murthy
2. Foundation Analysisa nd Design by JosephBowles
3. Foundation Engineering by P.C.Varghese
4. Pile Foundation in Engineering Practice by Shamsher Prakash & Hari D. Sharma
5. Advance Soil Mechanics byBrajaM Das

R.K.D.F. UNIVERSITY, BHOPAL

M.TECH **(Construction Technology and Management)**

Second Year (**Semester – 3**)

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Construction Technology and Managment)	**CONSTRUCTION PROJECT AND MANAGEMENT**	**MTCTM–3002**	**3L-1T-0P**	**4**

Course objectives:

This course will enable students to

- Understand the various management techniques for successful completion of construction projects.
- Understand the effect of management for project organization,

Unit-1

Introduction: Construction Projects- Concept, Project Categories, Characteristic of projects, project life cycle phase. Project Management- Project Management Function, Role of Project Manager. Organizing For Construction - Principles of organization, type of organization structure

.Unit-2

Project Feasibility Reports: Introduction, Significance in feasibility report- Technical analysis, Financial analysis, Economic analysis, Ecological analysis, Flow diagram for feasibility study of a project. Project planning Scope: Planning Process, Objectives, Types of Project plans, Resource Planning Process.

Unit-3

Bar Charts, Work Breakdown Structure, Time estimates, Applications of CPM and PERT A-O-N Network-Logic and Precedence diagrams, advantages, Drawing A-O-N network from A-O-A network and related problems

Unit-4

Time Cost relationship: Direct and indirect cost, step in optimization of cost, related problem. Allocation of resources: Histogram, Resource smoothening, Resource leveling and related problem. Project updating using CPM network and related numerical problems.

Unit-5

Scheduling, Monitoring and Updating. Line of Balance Scheduling. Resource Planning-Leveling and Allocation. Introduction to Building Information Model (BIM).

REFERENCES:

1. Chitkara, K.K. “Construction Project Management: Planning, Scheduling and Control”, Tata McGraw-Hill Publishing Company, New Delhi, 1998.

2. Choudhury S , “Project Management”, McGraw-Hill Publishing Company, New Delhi, 1988.

3. Chris Hendrickson and Tung Au, “Project Management for Construction – Fundamental Concepts for Owners, Engineers, Architects and Builders”, Prentice Hall, Pittsburgh, 2000.

4. Srinath L.S, “PERT and CPM”, East West Press Pvt Ltd New Delhi.

5. Frank Harris and Roland McCaffer, “Modern Construction Management”- 4th Ed Blackwell Science Ltd.

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Construction Technology and Managment)	**Seminar**	**MTCTM-3003**	**0L-0T-4P**	**4**

Students have to give a seminar both in written and oral formats based on the following:

- Recent advancement in the field of Civil Engineering practice and research.
- Scope of further technological advancement in various fields of Civil Engineering.
- Various new techniques, field problems and innovative ideas
- Any other relevant topics

R.K.D.F. UNIVERSITY, BHOPAL

M.TECH **(Construction Technology and Management)**

Second Year (**Semester – 3)**

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Construction Technology and Managment)	**Dissertation-1**	**MTCTM-3004**	**0L-0T-8P**	**8**

- Identify structural engineering problems reviewing available literature.
- Identify appropriate techniques to analyze complex structural systems.
- Apply engineering and management principles through efficient handling of project